

TECHNICAL SPECIFICATION

for the development of software for an electronic system implementing the service “Habit Tracker”

APPROVED BY

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1. List of conventional abbreviations

Term	Meaning
HABIT TRACKER (SYSTEM)	A web platform for planning, tracking, and analyzing users' daily habits and goals.
ADMINISTRATOR (ADMIN)	A privileged user who performs technical administration of the system, manages user accounts, and maintains global category directories.
USER	An authenticated person who uses the system to create, edit, and track their own habits and to view personal statistics.
PC (PERSONAL CABINET)	A personalized web page / interface (User panel or Admin panel) for interacting with the system's functionality according to access rights.
HABIT	The core entity in the system. It describes a regular action (for example, "Hydration routine", "Reading for 20 minutes") that has a completion schedule, a goal, and a status.
CONTROL PANEL (DASHBOARD)	The main graphical user interface for tracking progress, marking habits completed for the current day, and viewing analytics.
CRUD	A set of basic data operations: Create, Read (view), Update, and Delete.
API	Application Programming Interface — a programmatic interface for application-level interaction.
DR (DISASTER RECOVERY)	Emergency restoration of operability (Disaster Recovery).
AS	Automated system.
DB	Database.
BT/BK	Business tasks / Business knowledge.

2. Purpose of the document

The task of this document is to provide a description of the functionality required to carry out the work on developing software for an electronic system that implements the "Habit Tracker" service.

2.1. Purpose of the development

To provide a description of the functional and technical requirements for developing the "Habit Tracker" web platform, which automates the process of forming users' beneficial habits.

2.2. Purpose of the system

The system is intended to help users with self-organization, achieving goals, and forming systematic daily actions, and also to support:

1. Maintaining a user database with secure storage of their personal data and settings.
2. Creating and flexibly configuring habits (name, category, frequency, reminders).
3. Recording daily completion of actions (logs) with the ability to track dynamics.
4. Displaying clear analytics and statistics (day series / "streaks", completion percentage, charts).
5. Providing administrative control over the system's content and users.

2.3. Description of the subject area

The subject area is personal time management, self-development, and productivity. The web platform allows a user to move away from paper notebooks and keep a systematic electronic record of daily rituals, with visualization of their own achievements.

2.4. Planned start and finish dates of the work

Network schedule and critical path of the project (01.09.2026 — 05.12.2026). Development of the system is divided into sequential-parallel stages:

1. Requirements analysis and technical specification (01.09–12.09): gathering requirements and approving the specification.
2. Design and UI (14.09–26.09): parallel database design and development of UI templates.
3. Modules and authorization (28.09–17.10): development of QES/USR authorization and business processes.
4. Electronic cabinets (12.10–07.11): programming of user cabinets.
5. Integration and configuration (09.11–21.11): combining services, logging, and monitoring.
6. Testing on DEV/STAGE (23.11–28.11): deployment and defect resolution.
7. Acceptance testing (30.11–05.12): final project delivery during the examination week.

Critical path (14 weeks): Requirements analysis → Database design → QES authorization → Electronic cabinets → Integration → Testing → Acceptance.

Any delay on this chain shifts project delivery beyond the examination week (05.12.2026).

3. System requirements

3.1. Functional requirements

The system shall implement the following core functionality:

1. Registration and authorization: creating a new account, secure sign-in with a login and password, sign-out, and role separation (User / Admin).
2. Habit management (CRUD).
 - a) C (Create): creating a new habit (name, description, category, frequency: daily / by days of the week, target indicator).
 - b) R (Read): viewing the list of one's habits for today, filtering by categories, viewing details and history.
 - c) U (Update): editing habit parameters and changing the completion status for a specific day (completed / not completed).
 - d) D (Delete): deleting or archiving habits that are no longer relevant.
3. User panel (user interface): viewing the dashboard for the current day, a completion calendar, and a block of overall statistics (successful-day streak, percentage of goals completed).
4. Admin panel (administrator interface): managing the user list (blocking, editing), managing shared habit categories (category CRUD), and viewing overall system statistics.
5. Database: centralized and reliable storage of data about users, habits, completion logs, and categories.

3.2. Non-functional requirements

1. Usability: an adaptive, modern, and intuitive web interface.
2. Security: hashing of passwords in the database (for example, using BCrypt or ASP.NET Core Identity) and protection against unauthorized access to other users' habits.
3. Reliability: preserving data integrity in the relational database while performing CRUD operations.

3.3. Business rules

1. Each user sees and manages only the habits they have created themselves.
2. A habit must be bound to one of the categories (for example: Sports, Health, Learning, Productivity).
3. A habit's completion status is recorded separately for each day.

4. An administrator cannot change users' personal habits, but has the right to delete accounts or content that violates the rules.
5. Deleting a user account results in deletion of all habits bound to that account and of their completion history (cascade deletion in the database).

3.4. System requirements

1. Compatibility of the client part with popular browsers (Google Chrome, Mozilla Firefox, Microsoft Edge, Safari).
2. A centralized relational database (MS SQL Server / MySQL / PostgreSQL).
3. Server environment: the web application is deployed on an internal server running Windows Server (a member of the local domain) using the IIS web server.
4. Network access: users access the platform through the internal corporate or university network (Intranet) via a local domain name (for example, <http://habits.local>), using a standard login-and-password form.

3.5. Technological requirements (technology stack)

1. Programming language: C#.
2. Platform: .NET / .NET Core.
3. Web framework: ASP.NET Core MVC (or ASP.NET Core Web API + SPA).
4. Database: MS SQL Server / MySQL (using Entity Framework Core).
5. IDE: Visual Studio / JetBrains Rider.
6. UI/UX design and front-end: HTML5, CSS3, JavaScript (Bootstrap / Tailwind CSS).

4. Users and roles

4.1. User role model

Role name	Description	Rights and capabilities in the system
Unauthorized visitor	Any Internet user who has opened the home page.	1) Viewing the landing / home page that describes the platform's capabilities. 2) Completing the registration procedure in the system. 3) Authorization (sign-in) to their account.
Authorized user (User)	An ordinary user who has passed authentication and has access to the User panel.	1) Managing their own data (CRUD for habits). 2) Marking habit completion on a daily basis. 3) Viewing their own analytics, calendar, and statistics. 4) Editing their own profile data (name, password, avatar).
Administrator (Admin)	A privileged staff member / system moderator who has access to the Admin panel.	1) Full control over users (viewing the list, blocking / deleting accounts). 2) Managing global directories of system habit categories (CRUD). 3) Viewing overall platform analytics (number of active users, total number of habits).

4.2. Use-case and role diagrams

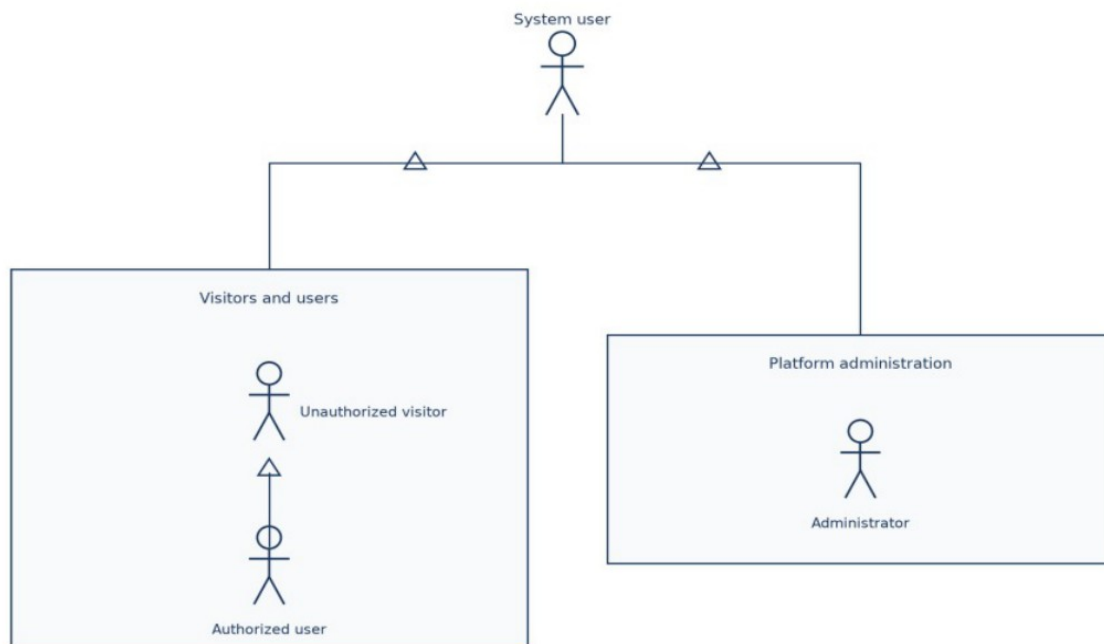
Figure 1. Role model of system users

Figure 2. Use-case diagram of the information system “Habit Tracker”

